PHASE 1 DOCUMENT

EduGenie: Google Gemini Powered Learning Assistant

Phase Name: Project Proposal and Requirement Analysis

Project Name	EduGenie: Google Gemini Powered Learning Assistant
Phase	Phase 1 – Proposal and Requirements
Project Type	AI-Powered Learning Application
Main Technology	Google Gemini API
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 1 of the project **EduGenie: Google Gemini Powered Learning Assistant**. The purpose of this phase is to define the project idea, identify the learning-related problem it addresses, establish objectives and scope, and collect the functional and non-functional requirements needed before system design and development begin.

2. Project Abstract

EduGenie is an AI-powered learning assistant that supports students with concept explanations, doubt clarification, quiz generation, revision planning, summaries, and project guidance. The system uses Google Gemini through a secure backend integration to understand student questions and generate helpful learning responses. EduGenie is intended to complement classroom learning by making academic support more accessible, personalized, and interactive.

3. Problem Identification

Students frequently need academic support outside classroom hours. They may face difficulty understanding complex concepts, finding suitable practice questions, preparing revision schedules, or receiving immediate guidance for project work. Faculty support is valuable but may not always be instantly available for every student. This creates a need for a learning platform that can provide on-demand assistance while encouraging students to verify important information with approved academic resources and faculty guidance.

4. Proposed Solution

EduGenie will provide a simple web-based interface where students can select a learning activity or type a question in natural language. The application will send the request to a secure backend service, which connects with Google Gemini. The AI response will be presented to the student in a clear format. Depending on the selected feature, the system can explain a topic, generate a quiz, create a study plan, summarize content, or provide project-documentation support.

5. Project Objectives

The project aims to provide accessible on-demand learning assistance; explain concepts at different levels of detail; generate topic-based quizzes and practice questions; support study planning and revision; assist students in organizing project ideas and documentation; promote independent learning; and use Google Gemini responsibly through secure API integration and suitable safety controls.

6. Scope of Phase 1

Phase 1 covers project planning and requirement analysis only. It defines the proposed EduGenie features, user roles, system boundaries, requirements, assumptions, and success criteria. It does not include user-interface implementation, database creation, Google Gemini API coding, deployment, or final testing. Those activities will be handled in later project phases.

7. Target Users

The primary users are students who need support for learning, revision, quizzes, and academic projects. Faculty members may use the system to share approved learning resources or suggest learning topics if this capability is enabled. Administrators manage accounts, subject categories,

system settings, and access permissions. The first implementation can prioritize the student role and add faculty and administrator features incrementally.

8. Functional Requirements

The system shall allow users to register and log in. The system shall allow students to enter questions in natural language. The system shall provide AI-generated explanations using Google Gemini. The system shall support quiz generation based on a selected topic. The system shall allow users to request study plans and revision support. The system shall provide project guidance such as title ideas, objectives, module suggestions, and documentation outlines. The system shall allow students to save useful responses or learning content when the feature is enabled. The system shall display appropriate error messages if the AI service is unavailable.

9. Non-Functional Requirements

EduGenie should provide a simple and responsive interface for desktop and mobile users. The system should protect user accounts and data using authentication and authorization. Google Gemini API credentials must be stored securely in the backend and must not be exposed in the frontend. The application should provide understandable error messages, maintain reasonable response times, and be designed for scalable usage. The system should follow privacy-aware practices and avoid unnecessary collection of personal or sensitive data.

10. System Boundaries

EduGenie provides learning assistance, not guaranteed authoritative answers. It does not replace teachers, official textbooks, institutional assessment rules, or professional services. The system should remind users to verify important academic information. The project will not use AI to make final academic decisions, grade students automatically without faculty oversight, or provide professional medical, legal, or emergency advice.

11. Assumptions and Dependencies

The project assumes that users have internet access and a supported web browser. It depends on access to a Google Gemini API service, a secure backend environment, and a chosen database or storage mechanism for accounts and optional learning history. Response quality may depend on the clarity of the student prompt and the availability of the external AI service.

12. Success Criteria

Phase 1 will be considered complete when the problem statement, objectives, project scope, user roles, functional requirements, non-functional requirements, constraints, and planned modules are documented and approved. The overall project will be successful when students can use EduGenie to receive clear learning support through a secure, usable, and responsible AI-assisted interface.

13. Planned Next Phase

The next phase will be **Phase 2: System Design and Architecture**. It will define the application modules, user-interface flow, data design, user roles, backend components, and secure Google Gemini integration flow based on the requirements collected in this phase.

14. Conclusion

Phase 1 establishes the foundation for EduGenie. The proposed application addresses the need for accessible and personalized academic support by combining a student-focused learning interface with Google Gemini. The documented requirements will guide the design, implementation, testing, and deployment activities in the upcoming phases.

Phase Approval

Prepared by	_____
Guide / Faculty	_____
Approved by	_____